

KEVIN RUSAGARA

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EDUCATION

Yale University

New Haven, CT (Aug 2023 - May 2027)

Bachelor of Science in Computer Science

Courses: Algorithms, Computer Architecture, OOP, Distributed Systems, Compilers, AI, Machine Learning

SKILLS AND INTERESTS

Programming Languages: Python, C/C++, Swift,Java, JavaScript, Racket, SQL, Bash

Libraries/Frameworks: FastAPI, Flask, Spring Boot, React.js, Next.js, Bootstrap, PyTorch, TensorFlow

Tools / Platforms: Git, AWS, Postman, Google Firebase, Linux, OpenCV, Captum, NumPy, Matplotlib, MS Excel

EXPERIENCE

Yale University & NSF | Software Developer

New Haven, CT | Aug 2025 – Present

• Building core software infrastructure for an open research ecosystem supporting hyperspectral imaging, including data pipelines and file converters for ENVI, BIL, BIP, BSQ, GeoTIFF, and PNG formats.

• Developing Assembler and Dashboard modules using React.js and JavaScript to enable scalable visualization, transformation, and management of multi-band spectral datasets.

• Collaborating with interdisciplinary research teams to standardize data workflows and improve reproducibility across imaging experiments.

UL Solutions | Software Engineer Intern

Fremont, CA | Jun 2025 – Aug 2025

• Developed Python-based automation tools to control RF test equipment and orchestrate SAR (Specific Absorption Rate) measurements, reducing manual setup time and increasing test repeatability.

• Partnered with RF and embedded systems engineers to debug test workflows, document compliance processes, and build scripts for firmware version tracking and reporting.

AI4ALL | AI Engineer Fellow

New Haven, CT | Sep 2024 – Apr 2025

• Designed and trained convolutional neural networks using the Labeled Faces in the Wild dataset (13,000+ images), with emphasis on model interpretability and evaluation.

• Applied Captum-based saliency maps and Grad-CAM visualizations to analyze model decision-making and improve transparency in ML systems.

Citadel Securities | Software Engineer Extern

New York, NY (Hybrid) | May 2024 – Aug 2024

• Built a Python and Flask-based web application to track expenses and simulate portfolio performance using real-time market data APIs.

• Evaluated and tested multiple trading strategies within a market simulation environment, gaining exposure to financial data pipelines and quantitative workflows.

Awesomity Lab | Backend Software Engineer Intern

Kigali, Rwanda | Jun 2024 – Aug 2024

• Developed the backend for Provigator, a production bug-tracking system used internally to manage and resolve 320+ software issues.

• Implemented RESTful APIs with FastAPI and SQLite, supporting full CRUD operations for users, projects, and clients.

• Integrated Google OAuth authentication to improve security and streamline user access.

PROJECTS

AI Agent Identity & RAG Gateway

Python, FastAPI, JWT, OAuth2, FAISS, LangChain, Docker, Git

• Designed and implemented a backend system that treats AI agents as first-class identities, enabling secure authentication, scoped authorization, and lifecycle management for agentic workflows.

• Built protected REST APIs using FastAPI and JWT-based access control, enforcing fine-grained permission scopes for agent execution and RAG queries.

• Implemented a RAG pipeline backed by FAISS vector search, supporting access-controlled document retrieval and tool-based agent execution with structured request logging.

Hyperspectral Imaging Ecosystem

React.js, JavaScript, ENVI, GeoTIFF, AWS, Git

• A Yale–NSF collaborative platform for hyperspectral imaging research, building tools that standardize and visualize multi-band spectral data.

- Developed core components for the Assembler and Dashboard, enabling researchers to convert and manage ENVI files into BIL, BIP, BSQ, GeoTIFF, and PNG formats.
- Engineered an intuitive web interface in React.js for efficient data exploration, improving usability and reproducibility across imaging workflows.

Facial Recognition with Saliency Maps CNN, PyTorch, Captum, OpenCV, Matplotlib, TensorFlow, Keras

- A facial recognition model under development using CNNs in PyTorch aiming to achieve over 95% accuracy, integrating Captum for gradient-based visualizations like saliency maps and Grad-CAM. Leverages OpenCV and NumPy for image preprocessing, with Matplotlib visualizing metrics and performance. Focuses on model interpretability to enhance understanding of how the model makes decisions.

Provigator API Python, FastAPI, SQLite, Google OAuth

- A FastAPI-based RESTful backend for a bug-tracking system at Awesomity Lab, using SQLite for centralized management of clients, projects, and users. Integrates Google OAuth for secure authentication and supports full CRUD operations. Tracked and resolved over 320 bugs, significantly improving software quality assurance at Awesomity Lab.

LEADERSHIP

Yale Computer Society | Mentor, Catalyst Program New Haven, CT | Jan 2025 – Present

- Mentored early-stage student founders in the Catalyst Program on product ideation, technical feasibility, and go-to-market strategy.
- Provided structured feedback on MVP development, user validation, and pitch preparation for demo days and funding opportunities.

Yale African Innovation Symposium | Chief Business Officer New Haven, CT | Aug 2024 – Present

- Led business strategy, fundraising, and sponsor partnerships for a pan-African innovation conference hosted at Yale University, convening students, founders, investors, policymakers, and technologists.
- Secured and managed \$100K+ across two consecutive symposium cycles through corporate sponsorships, institutional grants, and philanthropic partnerships.